ForgeAI
ENGINEERING INTELLIGENCE

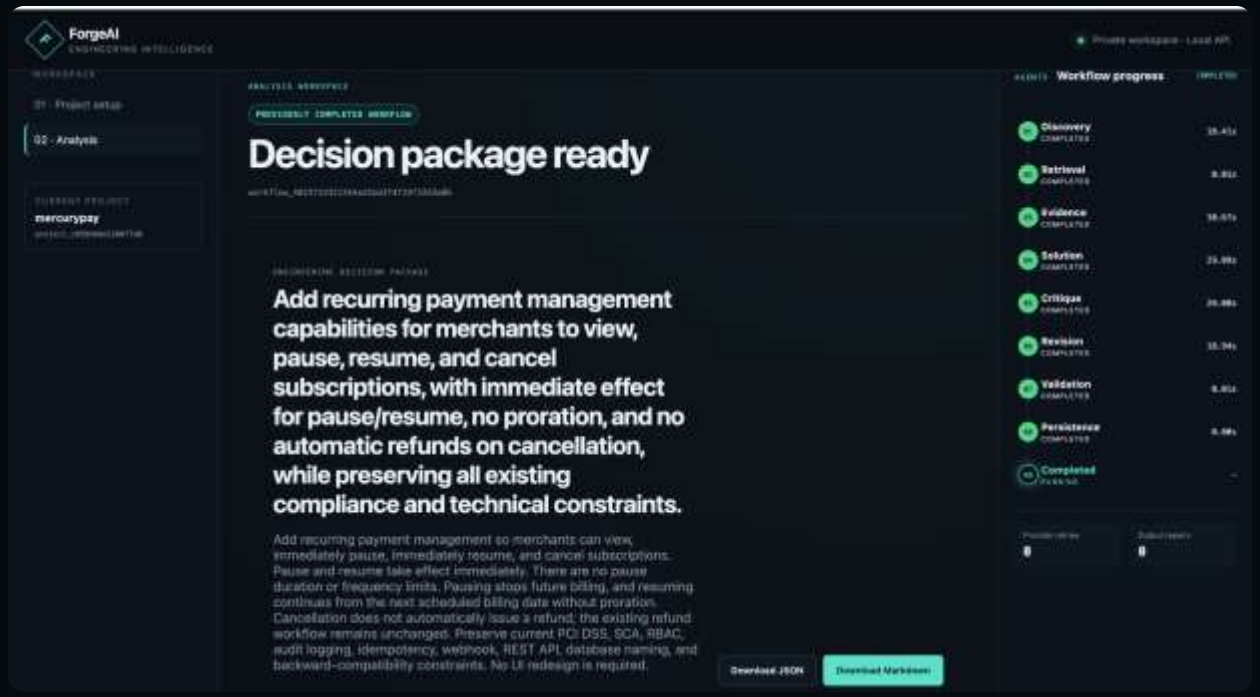
AMD AI DEVMASTER · TRACK 2

ForgeAI

Private, evidence-backed engineering decisions through deterministic multi-agent analysis.

VERIFIED AMD RADEON CLOUD INFERENCE

5	106s	0
bounded reasoning roles	verified AMD workflow	retries / repairs



ABHISHEK HERBERT SAMUEL

SOLO PARTICIPANT

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PROBLEM

Engineering review still lives between scattered facts and ambiguous asks.

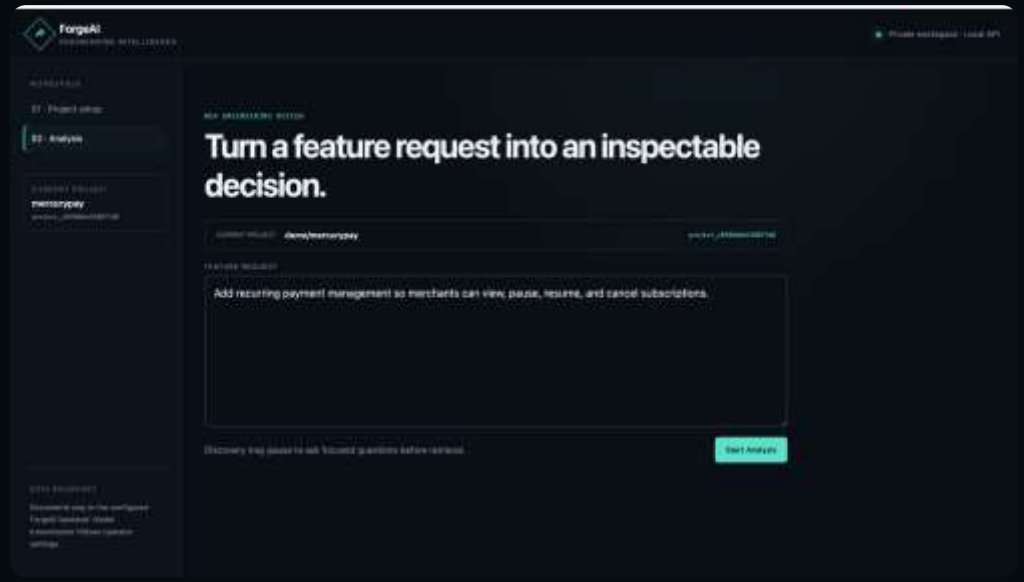
Feature requests arrive incomplete. Local architecture, compliance constraints, API contracts, and past decisions are scattered. Generic chatbots often produce plausible output without provenance or durable review state.

Ambiguous request
Different teams implement different interpretations

Scattered local knowledge
ADRs, policies and schemas are overlooked

Opaque AI output
Recommendations cannot be audited

Unchallenged proposal
Security and operational risks survive review



ForgeAI turns the pre-coding decision gap into a bounded, inspectable workflow.

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SOLUTION OVERVIEW

Feature request → Decision package.

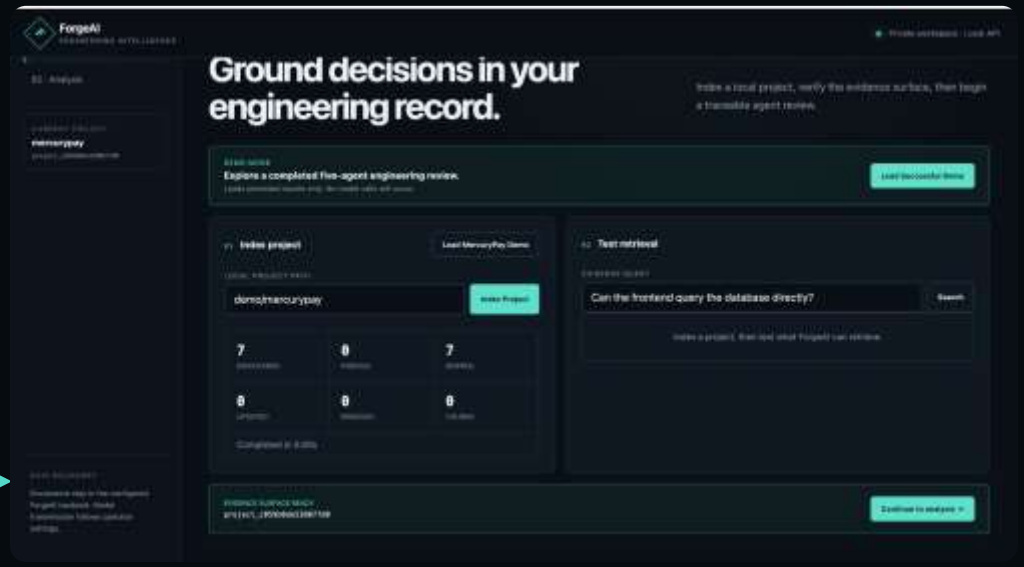
A fixed agent workflow produces an artifact that can be reviewed, exported, and revisited.

Request → **Local docs** → **Agents** → Validated decision

Ambiguous feature ask

MercuryPay evidence

Discovery → Revision



JSON — machine-readable export

MD — human-readable export

EDP — decision package

SYSTEM ARCHITECTURE

One engine. Two transports.

Explicit boundaries: transport code never owns workflow logic.

- Provider-independent domain invariants
- Application-owned transitions and validation
- Adapters for parsing, retrieval, SQLite and inference
- Frontend consumes API, never database internals



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FIVE-AGENT WORKFLOW

Fixed state machine. Bounded agents.

Human checkpoint

Discovery pauses before retrieval, asks ordered questions, and resumes without rerunning earlier reasoning.



- Typed input/output contracts
- Invalid output cannot advance
- Events, timings and provider metadata are persisted
- Critic findings receive exactly one Revision disposition

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TRUST ARCHITECTURE

The model chooses. ForgeAI proves.

Reasoning models are not asked to reproduce hashes, paths, excerpts, line ranges or canonical IDs.



Trusted public EvidenceOutput

IDs, paths, lines, excerpt and hash copied from retrieval

- Model never supplies provenance
- Unknown or duplicate labels rejected
- Raw model outputs not persisted

AMD INTEGRATION

Verified on AMD Radeon Cloud.

The successful live workflow used AMD Radeon Cloud's shared DeepSeek-V4-Flash endpoint through a provider-neutral OpenAI-compatible adapter.

Model	DeepSeek-V4-Flash
Served as	radeon-deepseek/DeepSeek-V4-Flash
Mode	json_object
Timeout	300s
Provider retries	0



106.29s
total command latency

20,556
model tokens

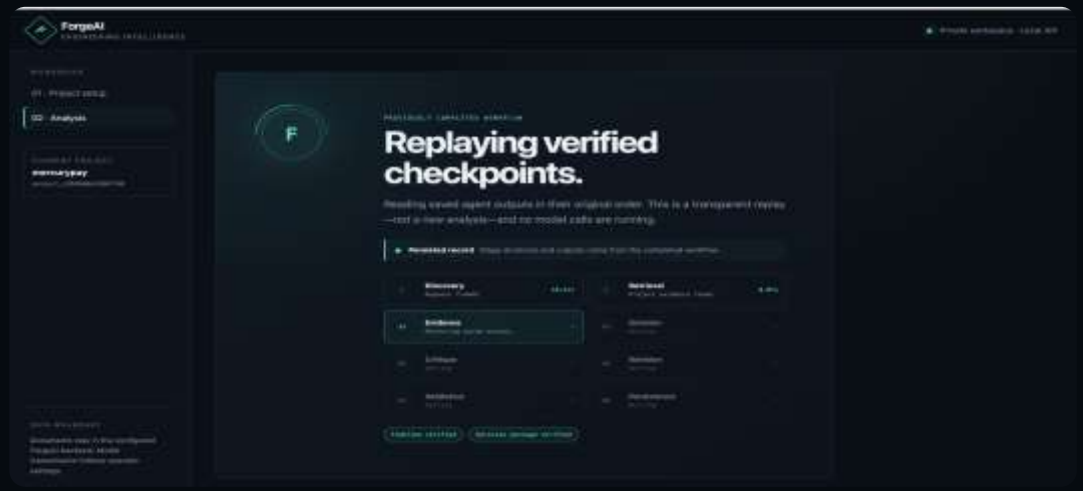
10/10
critic dispositions accepted

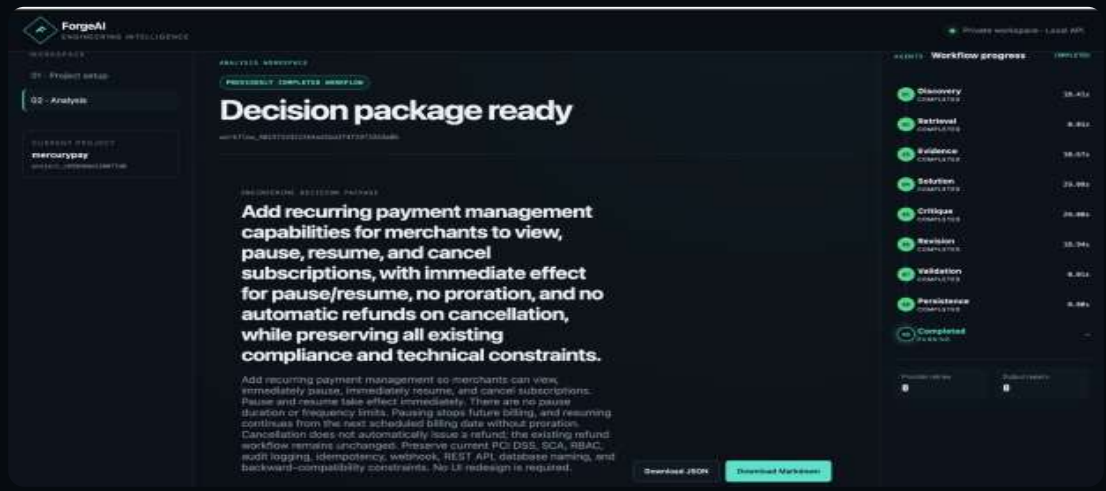
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PRODUCT EXPERIENCE

A workspace built for trust, not theatre.

Live analyses and completed workflow inspection are separated. The demo replay is explicitly labeled and uses persisted outputs only.





- Visible stage timings
- Safe failure states
- JSON + Markdown exports
- Accessible controls

ENGINEERING QUALITY

Production discipline inside an MVP.

156	16	93%	0
backend tests passed	frontend tests passed	backend coverage	live output repairs

- Ruff format/check
- MyPy strict typing
- Vitest + accessibility assertions
- OpenAPI 3.1 contract
- SQLite WAL + migrations
- Credential-free mock provider

Failure-safe design

- Exactly one bounded repair
- Sanitized diagnostics only
- No raw provider output persisted
- Invalid model output cannot silently advance
- No model-controlled commands or repo mutation

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ROADMAP

From verified shared inference to dedicated Radeon deployment.

ForgeAI

Engineering intelligence for defensible decisions.

Now

Local modular monolith + AMD shared DeepSeek API

→

Next

Dedicated Radeon Cloud vLLM endpoint + ROCm telemetry

→

Scale

Auth, project history, GitHub/Jira, vector retrieval, workers